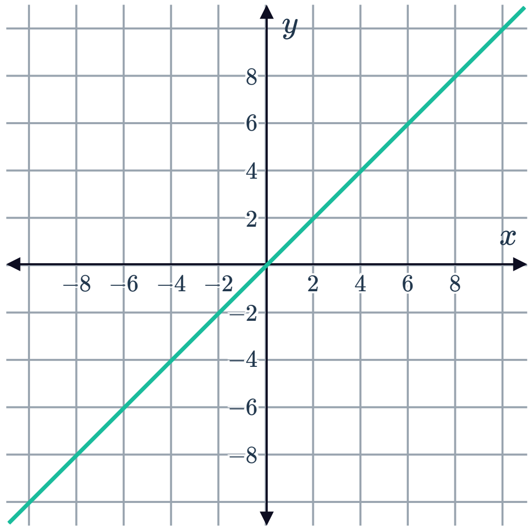
Gradient function graphs

1 Consider the graph of $y = x$:

a Find the gradient of the line at $x = 4$.

b Find the gradient at any value of x.

c What can be said about the gradient of a linear function?



2 Consider the graph of $y = x^2$ and its gradient function $y = 2x$:

a What can be said of the sign of the gradient function when $y = x^2$ is increasing or decreasing?

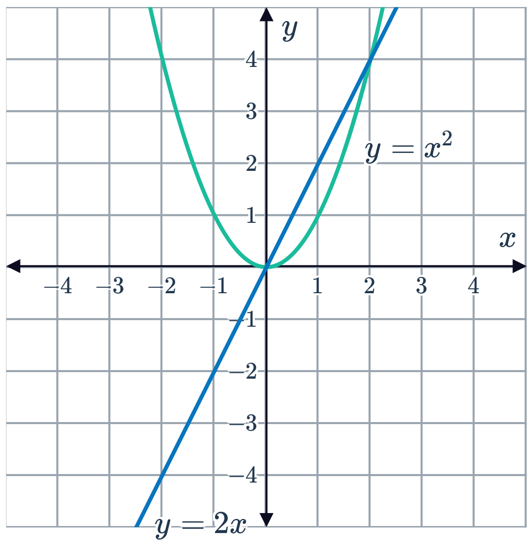
b For $x > 0$, is the gradient of the tangent positive or negative?

c For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?

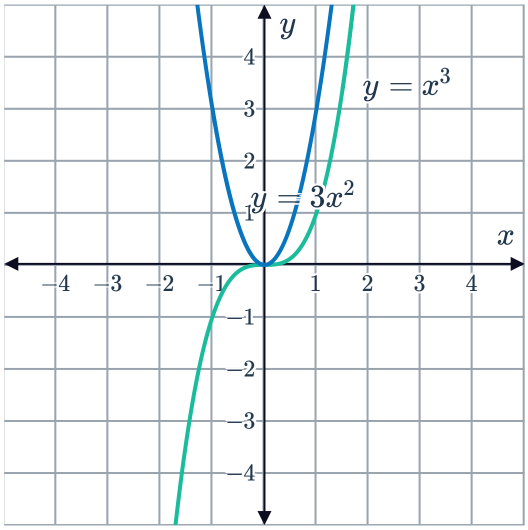
d For $x < 0$, is the gradient of the tangent positive or negative?

e For $x < 0$, as the value of x increases how does the gradient of the tangent line change?

f For $y = x^2$, the gradient of the tangent line changes at a constant rate. What type of function is the derivative of $y = x^2$?



3 Consider the graph of $y = x^3$ and its gradi[redacted]nction $y = 3x^2$.



a For $x > 0$, is the gradient of the tangent positive or negative?

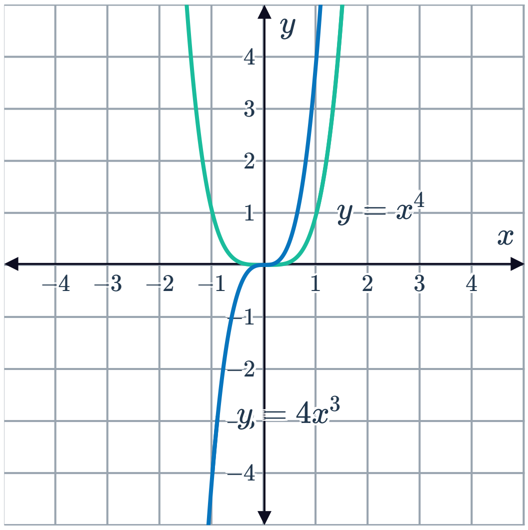
b For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?

c For $x < 0$, is the gradient of the tangent positive or negative?

d For $x < 0$, as the value of x increases how does the gradient of the tangent line change?

e For $y = x^3$, the gradient of the tangent line first decreases at a decreasing rate, then increases at an increasing rate. What type of function is the derivative of $y = x^3$?

4 Consider the graph of $y = x^4$ and its gradient function $y = 4x^3$.



a For $x > 0$, is the gradient of the tangent positive or negative?

b For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?

c For $x < 0$, is the gradient of the tangent positive or negative?

d For $x < 0$, as the value of x increases how does the gradient of the tangent line change?

e For $y = x^4$, the gradient of the tangent line is increasing, first at a decreasing rate and then at an increasing rate. What type of function is the derivative of $y = x^4$?

5 Consider the functions $f(x) = x^5$ and $g(x) = x^4$.

a Sketch the graph of $f(x)$ and its derivative.

b Sketch the graph of $g(x)$ and its derivative.

c If the degree of a function is even, will the degree of its derivative function be odd or even?

d If the degree of a function is odd, will the degree of its derivative function be odd or

Algebraic differentiation

6

a For each of the following functions, state the degree of the gradient function:

i $y = x^2$ ii $y = x^3$ iii $y = x^4$

b Hence, state the degree of the derivative of a polynomial function of degree n.

7 Find the derivative of the following functions with respect to x:

a $y = x^7$ b $y = x^9$ c $y = x^5$ d $y = -6$

Applications

8 Find the gradient of $f(x) = x^4$ at $x = 2$. Denote this gradient by $f'(2)$.

9 Consider the function $f(x) = x^9$.

a Find the gradient of the tangent to the function at $x = 1$.

b Find the equation of the tangent to the function at $x = 1$.

10 David draws the graphs of x^2, x^3, x^4 and x^5 and draws the tangents to each one at the point where $x = 1$.

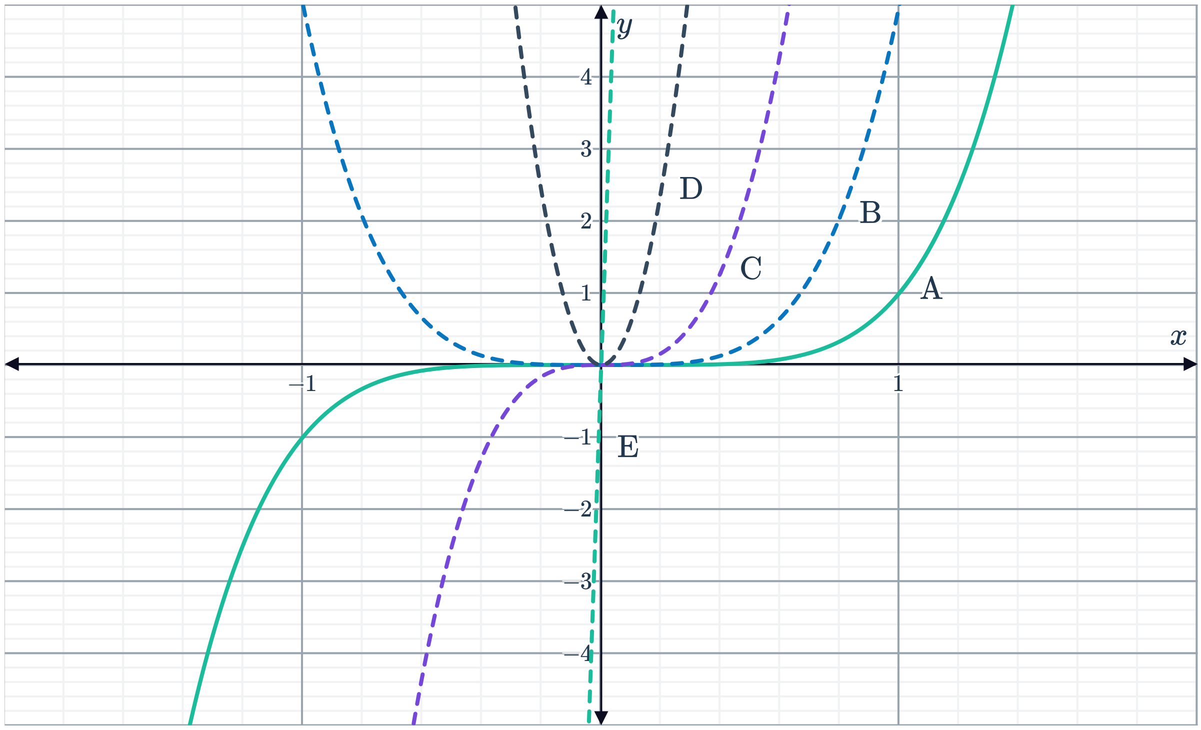
a David then notices where each of the tangents cut the y-axis and records this in the table below:

Graph	y-intercept of the tangent line	Gradient of the tangent
$y = x^2$	$(0, -1)$	
$y = x^3$	$(0, -2)$	
$y = x^4$	$(0, -3)$	
$y = x^5$	$(0, -4)$	

Complete the table by calculating the gradient of each of the tangents at $x = 1$.

b Following the pattern in the table, what would be the gradient of the tangent to the graph of $y = x^n$ at the point where $x = 1$?

11 The graph of $y = x^5$ is shown below labell[redacted] A. Fiona then graphs the derivative of the function, labelling it as B. She then finds the derivative of graph B to get graph C, then differentiates again to get graph D and differentiates again to get graph E.



State whether the following statements are true about this sequence of derivatives:

a The derivative of a function is always positive when the function is negative, and negative when the function is positive.

b Each graph is a function of the form ax^n.

c For any value of x, the value of the derivative will always be greater than the value of the function.

d The degree of the derivative is always different to the degree of the function.